Bayesian Optimisation of a Bio-Based Polyester Particulate-Reinforced Composite for Cured Compressive Properties

7th July 2025

Thomas Højlund-Dodd, James Elliott

Abstract

This paper will focus on particular group of thermosets, namely, polyester resins. These liquid resins generally cured to form the matrix that surrounds reinforcing materials within the structure of composite materials such as glass-reinforced plastic. This paper initially elucidates the method by which an entirely bio-based polyester resin was manufactured, from simple organic acids and alcohols. Subsequently, a description is provided of how Bayesian optimisation was applied to improve mechanical properties of interest by using a number of predictors to tune the sample compositions. In the multi-objective setting, it was found that these materials could have their yield strength and Young's modulus improved upon up to a putative global optimum of 165 MPa and 5,458 MPa respectively. In the single-objective setting, an experiment succeeded in developing the Young's modulus of these bio-based materials up to 5,012 MPa which equated to 69% the best modulus obtained from an identical optimisation of the equivalent 'conventional' styrene-based material.

1 Introduction

Plastics encompass a plethora of polymeric materials which provide a wide-range of desirable material properties; and given this, have seen use in a variety of applications, spanning both commodity and engineering settings [30]. Since the 1950s, the annual mass of plastics manufactured has grown from 2 Mt yr^{-1} [29] to around 460 Mt yr^{-1} in 2019 [20]; naturally, plastic waste tracked this rise. In 2019, 353 Mt of plastic waste was generated globally, with 49% landfilled, 22% mismanaged, 29% incinerated, and 9% recycled [20]. Mismanaged plastics may degrade and pollute watercourses with microplastics [32], incinerated plastics provide a means by which fossil carbon may reach the atmosphere [22], and sluggish development of recycling globally makes this the least likely fate of waste plastic [20]; these are a subset of the many challenges born out of the use of this family of materials.

Since around 85% of virgin plastic manufactured is thermoplastic they have generally dominated plastic sustainability literature, with papers focussed upon improvement of recycling methods [35], mitigation of microplastic waste [51], and biodegradability [50]. Despite their proportional minority, this paper focusses upon the group of plastics called thermosets- plastics characterised by cross-linked network structures, high dimensional stability, and chemical corrosion resistance [45]. It is generally impractical to depolymerise cross-linked thermoset materials which has lead to the common practice of either landfilling or incinerating these materials at end-of-life [75, 11, 73]. Contemporary European legislation has prioritised incineration ahead of landfill for composite materials with thermosetting matrices, and incoming legislation (e.g. Directives '99/31/EC' and '2000/53/EC') is likely to increase the proportion incinerated [57]. As the majority of thermoset plastics are derived from fossil sources of carbon, their combustion releases the greenhouse gas CO_2 into the atmosphere, perturbing the short-term carbon cycle [42]. This research directly treated this challenge through the application of Bayesian optimisation in the development of strength in bio-based alternatives to fossil incumbents.

Many thermosets exist including (but not limited to) phenolic resins, unsaturated polyester resins, epoxy resins, and polyurethanes [31]. A prominent group are the unsaturated polyester resins, which are routinely used in the manufacture of composite components; these resins exhibit desirable properties including ease of handling in liquid form, rapid curing, and excellent dimensional stability [4, 39]. The simplest form of conventional unsaturated polyester resin would be that formed from the mixture of poly(propylene glycol fumarate) copolymer (polymerised from propylene glycol and maleic anhydride) with a styrene reactive diluent [1]. Unfortunately, all monomers associated with conventional unsaturated polyester resins are fossil-derived products: propylene glycol is industrially manufactured from propylene oxide- a downstream chemical product of propane extracted from natural gas [42]. Maleic anhydride is synthesised by catalytic oxidation of the chemical butane [26]- this derived from fractional distillation of petroleum crude [37]. And styrene is produced from the dehydrogenation of ethylbenzene- this being the product of a reaction between benzene and ethylene, both of which are fossil-fuel derived feedstocks [40, 42].

Recently, literature has explored development of bio-based synthesis of these monomers; it has been shown that propylene glycol can be synthesised from the biodiesel-manufacturing byproduct glycerol via hydrogenolysis [19, 59], or from the hydrogenation of the sugar-fermentation derived product lactic acid [7, 77]. Meanwhile, bio-based maleic anhydride can be synthesised from either oxidehydration of n-butanol (from biomass fermentation of C5 and C6 sugars) [56] or catalytic oxidation of furfural (from the dehydration of C6 sugars) [44, 72]. Even bio-based styrene has been successfully manufactured via a biosynthetic route from glucose [48].

This paper focusses upon bio-based unsaturated polyester resins based on itaconic acid, a simple bio-based difunctional carboxylic acid manufactured by fungal fermentation of sugars by fungi such as *Aspergillus terreus* [5]. Itaconic acid's dicarboxyl and monoalkene functionality allows it to directly substitute maleic acid during polycondensation to form an unsaturated polyester [6]. Furthermore, itaconic acid can be synthesised into a variety of diesters (the family of diitaconates) which exhibit properties appropriate of a reactive diluent (i.e. radical initiated homopolymerisation and a low viscosity), allowing them to replace styrene in an eventual unsaturated polyester resin [66]. Bio-based itaconic acid has the advantage of being more directly synthesised from sugars than maleic acid, whilst diitaconates are a less volatile, less harmful and less flammable alternative reactive diluent to bio-based styrene.

Literature has explored catalytic influence upon polycondensation of itaconic acid within eventual unsaturated polyesters [62], synthesis of partially bio-based resins incorporating itaconic acid based unsaturated polyester but retaining styrene as the reactive diluent [14], as well as fully bio-based resins using itaconic acid based unsaturated polyester in tandem with diitaconates as reactive diluents [54, 53]. As an aside, literature has also considered partially bio-based resins incorporating itaconic acid based unsaturated polyesters together with the cross-linking compatible methacrylate class of reactive diluents [67, 18]. Although traditionally fossil-fuel derived, recent work has seen bio-based methyl methacrylate synthesised from itaconic acid, which would unlock fully bio-based resins based on methacrylate reactive diluents in future [52, 9]. Diitaconates were nontheless focussed upon in this research (despite inferior homopolymerisation rates [66] and viscosity characteristics [67]) ahead of methacrylates on account of their relatively higher strengths as cured thermosets [67], safety as reactive diluents (lower toxicity and flammability) [41], and environmentally-benign nature as a nonvolatile organic compound [67, 74].

On account of the temporal and financial expense of synthesising the chemicals concerned, the sample-efficient technique of Bayesian optimisation was used to facilitate the development of strength in these bio-based materials [27]. Literature describes how this technique has been used to successfully accelerate optimisation in a variety of domains: improving synthesis efficiency in organic chemistry [64], reducing phenotyping costs during crop breeding in plant science [71], and tuning microfluidic device setup for polymer fibre synthesis in materials science [43].

In polymer science, Bayesian optimisation has been deployed in both purely computational domains (e.g. optimisation of thermoplastic repeat units for improved glass transition temperatures in a simulated parameter space [49]) as well as in applied practice (e.g. improving compositional homogeneity during co-polymerisation of styrene and methyl methacrylate [70]). Various thermosets have been practically optimised using Bayesian optimisation: Benzoxazine-based thermosets have been optimised with respect to six thermomechanical properties [46], polycyanurates have had three thermal/mechanical/hygroscopic properties improved upon [78], and epoxy thermosets based on amine curing agents have been optimised with respect to a trio of mechanical properties using a dual-driver method [79]. In the more specific domain of practical Bayesian optimisation of bio-based thermosets, literature has been dominated by epoxy-based chemistries; Albuquerque et al. successfully improved glass transition temperatures in bio-based epoxy thermosets derived from amino acid based resin systems [2], whilst Rothenhäusler et al. explored this class of thermosets further by taking into account both cost and bio-based content [60]. Bayesian optimisation has seen practical application in the development of thermoset composites with a paper by Schoenholz et al. considering unidirectional carbon fibre prepreg based on epoxy resin; here, both the layup and curing cycle variables were optimised to reduce the process-induced deformation in final composite parts [61]. Despite this, no literature exists concerning the practical application of Bayesian optimisation in the development of properties in polyester-based thermosets, their bio-based analogues, or their derived composites. In applying Bayesian optimisation for the development of strength in simple composites derived from itaconate-based thermoset chemistries, this paper addresses this gap in the literature.

2 Method

Optimisation experiments involved both the practical methods associated with preparation and characterisation of thermoset composite samples as well as computational techniques developed to carry out Bayesian optimisation based on data obtained. Furthermore, synthetic routines were developed for producing any materials not readily available commercially. Initially, this section will present practical methods including synthesis of itaconic acid polyester and diesters, thermoset composite preparation, and characterisation techniques employed. Subsequently, computational method by which Bayesian optimisation was implemented will be elucidated.

2.1 Materials

The propylene glycol (99 %), itaconic acid (99 %), anhydrous calcium sulphate (99 %), and hydroquinone (99 %) were supplied by "Thermo Scientific Chemicals", a division of *Thermo Fisher Scientific Inc.* based in Waltham, Massachusetts, U.S. The toluene (99.8 %), dried magnesium sulphate (99 %), sodium chloride (99.5 %), and dichloromethane (95 %) were supplied by “Fisher Chemicals”, a division of *Thermo Fisher Scientific Inc.* The mixed xylenes (97 %) were supplied by “Alfa Aesar” a division of *Thermo Fisher Scientific Inc.* The methanol (99.9 %) was supplied by “Honeywell Riedel-de Haën”, a brand under *Honeywell International Inc.* based in Charlotte, North Carolina, U.S. The p-toluenesulphonic acid monohydrate (98 %) and chloroform-d (99.8 %) were supplied by “Sigma-Aldrich”, a subsidiary of *Merck KGaA* based in Darmstadt, Germany. Dibenzoyl peroxide (20 %) filled with calcium sulphate was supplied by *Nouryon Specialty Chemicals B.V.* based in Amsterdam, The Netherlands. Sodium bicarbonate (99 %) was supplied by *Fluorochem Ltd* based in Glossop, Derbyshire, U.K. Phenolphthalein solution (1 % w/w in ethanol) and potassium hydroxide (0.56 % w/w in methanol) were supplied by *Scientific Laboratory Supplies Ltd* based in the East Riding of Yorkshire, U.K.

2.2 Background

This research centred on the optimisation of a simple bio-based composite comprised of a thermosetting polymer matrix and a particulate reinforcing material.

Thermoset Polymer Matrix The thermosetting matrix was based on a crosslinked unsaturated polyester system; these rely upon a combination of condensation and addition polymer. The condensation polymer is typically an unsaturated linear polyester formed from the copolymerisation of diols and diacids, with at least one monomer having an unsaturated double bond in its structure which is retained after copolymerisation- this copolymer is referred to as an unsaturated polyester [63]. Whilst the addition polymer is typically generated through a radical-addition homopolymerisation reaction, where a monomer which had been acting as an unsaturated polyester solvent reacts and crosslinks individual unsaturated polyester chains- these monomers are therefore referred to as reactive diluents [12]. The combination of unsaturated polyester and reactive diluent is known as an unsaturated polyester resin; a viscous liquid, which, on addition of radical-initiating chemicals, can rapidly form a rigid thermoset material.

Previous work by Panic et al. found dimethyl itaconate to be the strongest contender of the diitaconates as a bio-based replacement for styrene in unsaturated polyester resins, given that resultant cured thermosets exhibited relatively higher crosslink densities (73 % the density of a styrene-based thermoset) [53]. This work therefore sought to replicate these unsaturated polyester resins, where the reactive diluent was pure dimethyl itaconate and the unsaturated polyester was poly(propylene itaconate).

Particulate Reinforcing Previous work by Kharas et al. made use of particulate calcium sulphate in the preparation of bioresorbable bone cement composites, where it’s content was known to effect the final thermosets’ strength and modulus [36]. This work therefore employed the readily-available anhydrous calcium sulphate in preparation as the particulate reinforcing element of the bio-based composites considered.

2.3 Synthesis of Diitaconate

Dimethyl itaconate was synthesised by esterification of itaconic acid with methanol by the Fischer method which had seen previous success in the literature [74, 15, 16]; this is presented in Figure 1 as a simplified reaction scheme. Method was adapted from the work of Trejbal et al. and involved the mixing of itaconic acid in an excess of methanol in a molar ratio of 1:3 [74]. P-toluenesulphonic acid was used as a het-

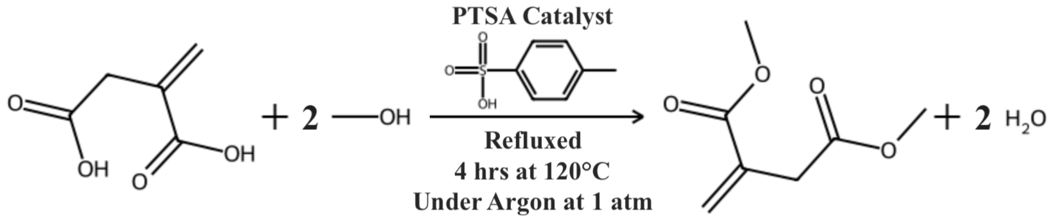

Figure 1: *Simple reaction scheme for synthesis of dimethyl itaconate.*

erogenous strong acid catalyst and made up 4 % w/w of the final reaction mixture. Figure 2 shows how this reaction mixture was placed in the lower round-bottomed flask (Fig.2D), and brought to a vigorous boil at around 130 °C. Vapours of both water and methanol passed up an insulated glass tube (Fig.2E) and were caught in a distillation column (Fig.2G) which was connected to a reboiler (Fig.2F) at it's base. Temperature at the column's top (Fig.2H) was retained at around 70 °C by adjusting the temperature of the reboiler (120 °C) and amount of insulation applied around the column; this ensured a steady stream of methanol-rich vapours passing through the Claisen head (Fig.2I) and re-condensing within the Liebig condenser (Fig.2J). Methanol-rich liquid was therewith returned to the lower round-bottomed flask via a dropping funnel (Fig.2K), whilst water was caught in the reboiler (Fig.2F), pushing the reversible esterification reaction toward the desired diester product.

Before initiating the reaction by starting both the hotplate stirrers (Fig.2A), an argon-rich atmosphere was generated by using a Schlenk line to cycle a vacuum (500 mbar) and argon gas (350 mL min^{-1}); this was to discourage unwanted oxidation reactions and reduce the chances of vapour ignition. During the reaction however a valve was opened to allow pressure equilibration between the reaction vessels and the atmosphere; mitigating the risk of explosion. Methanol cycling was halted 3.5 h into the reaction by closing the dropping funnel's tap (Fig.2K), after which, all heating was ceased on reaching 4 h.

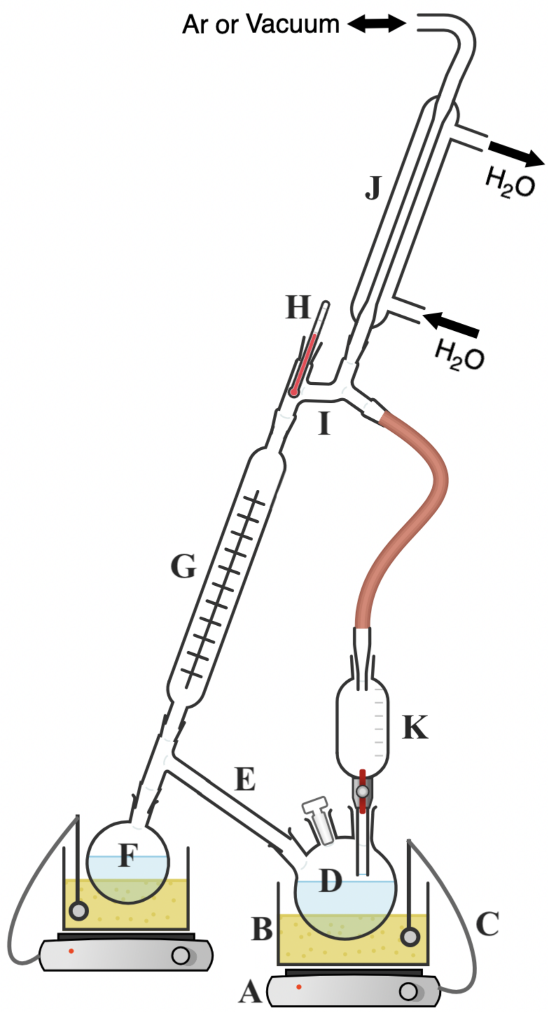

Figure 2: *Apparatus for synthesis of dimethyl itaconate [10]. (A) Hotplate stirrer (B) Sand bath (C) Temperature controller probe (D) Three-necked round-bottomed flask containing the reaction mixture (E) Insulated glass tube (F) Single-necked round-bottomed flask acting as a distillation column reboiler containing water (G) Distillation column (H) Thermometer (I) Four-way Claisen head (J) Liebig condenser (K) Dropping funnel for recycled methanol*

Workup was carried out in toluene; trial-and-error experimentation found this to be suitable for extraction of dimethyl itaconate. Final product mixture retrieved from the round-bottomed flask (Fig.2D) was placed in a separating funnel with toluene in a 1:3 volume ratio. Separating funnel contents was washed with water thrice, saturated sodium bicarbonate solution twice, and saturated sodium chloride once; each washing step involved a volume ratio of 2:1 between the previously obtained organic fraction and the next volume of washing liquid used. A gravity filtration step was carried out between the water and sodium bicarbonate washing steps to remove insolubles. Following washing, the organic fraction was removed to an Erlenmeyer flask and dried using anhydrous magnesium sulphate; thereafter removed using gravity filtration. Dry organics were placed in a rotary evaporator and heated at 40 °C whilst the pressure was dropped from an initial 70 mbar to 20 mbar over an average period of 8 h; this removed all but a small mass of toluene and water from the liquid product. From this liquid, dimethyl itaconate was crystallised out at room temperature by addition of a nucleating agent. The nucleating agent was crystalline dimethyl itaconate; serendipitously, storage of a particularly pure batch of rotary evaporation product in a fridge at 3 °C lead to the spontaneous nucleation required for freezing- generating the first crystalline sample of dimethyl itaconate. The above method afforded a 45 % yield although higher values may be possible if an extended period of boiling were adopted.

2.4 Diitaconate Characterisation

Characterisation techniques were used to monitor the reaction progress, identify chemicals, and ascertain the product purity; these included thin layer chromatography, Fourier-transform infrared spectroscopy, and nuclear magnetic resonance spectroscopy.

Initial impressions of purity were garnered from spotting chemicals onto thin layer chromatography plates (polyester with a layer of silica gel and ultraviolet indicator; 6 mm pore diameter and 0.2 mm layer thickness) and using a mixture of dichloromethane and methanol (2.5:7.5 v/v respectively) as the mobile phase. On completion, spot(s) could be observed under ultraviolet light (254 nm wavelength) and their retardation factors (R_F) calculated based on equation 1 relating this metric with the distance migrated by the substance in question (b in mm) and that travelled by the mobile phase (a in mm) [23].

$$R_F = \frac{b}{a} \tag{1}$$

Having solvated and diluted pure itaconic acid to a 1.25 % w/w solution (using stock mobile phase), it was spotted on a plate- this produced a single spot with an R_F value of 0.79. Meanwhile, having spotted a 5 % w/w solution of final crystalline product (expected to be dimethyl itaconate) a single spot with an R_F of 0.91 was obtained. These results provided benchmarks by which reaction progress could be gauged as well as supporting the hypothesis that these two crystalline materials were generally quite pure, although we could not rule out the presence of non-fluorescent or trace chemicals using this technique. Samples taken periodically from the reaction vessel (Fig.2D) during the course of an esterification reaction revealed a change in composition: the first sample (retrieved after 15min of heating) presented a single spot with an R_F value of 0.81, whilst a final sample (retrieved after 4 h) presented two spots with R_F values of 0.90 and 0.81 respectively. Given the similarity of these samples with those of the crystalline reactant and product analysed previously, these results indicate the partial conversion of the reactant itaconic acid into the desired product dimethyl itaconate.

Vibrational spectra were obtained using a Nicolet iS50 Fourier-transform infrared spectrometer using the attenuated total reflectance technique; the spectral range between 400-4000 cm^{-1} was investigated at a resolution of 0.09 cm^{-1}. Figure 3 presents the spectra obtained for the pure reactant (itaconic acid) and compares it with that of the product isolated using the rotary evaporator; this diagram supports the hypothesis that the product was succesfully esterified to form the product dimethyl itaconate in a number of ways: Firstly, the broad peak characteristic of O-H bond stretching commonly located between 2500-3200 cm^{-1} is clearly present for the itaconic acid but not for the product (Fig.3A), suggesting the loss of O-H bonds and therewith the carboxyl groups associated with either the initial reactant (itaconic acid) or the intermediate product (monoethyl itaconate). Secondly, strong peaks are observed around 1710 cm^{-1} (Fig.3B&3D) which is commonly indicative of C=O bond stretching (which occurs between 1640-1760 cm^{-1}); this would be consistent with C=O bond retention which would occur during succesful esterification.

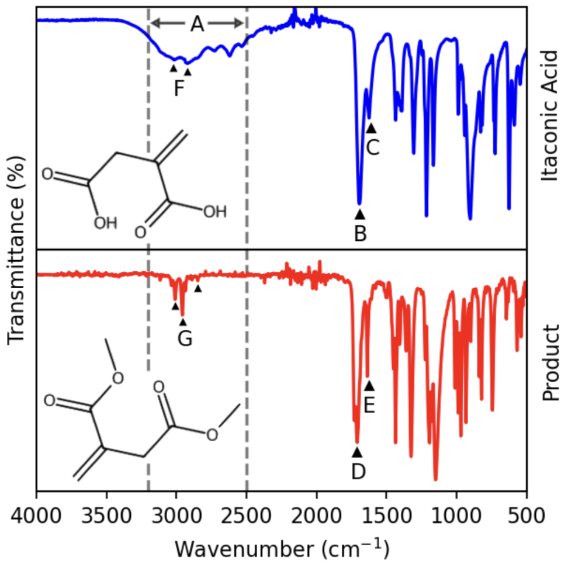

Figure 3: *FT-IR spectra of reactant and product. (A) 2500-3200* cm^{-1}*. (B) 1700* cm^{-1}*. (C) 1630* cm^{-1}*. (D) 1720* cm^{-1}*. (E) 1640* cm^{-1}*. (F) 2950 & 3050* cm^{-1}*. (G) 2875, 2980 & 3035* cm^{-1}*.*

Furthermore, it may be noted that peaks associated with carboxylic C=O bonds occur between 1640-1730 cm^{-1} whilst ester C=O peaks may occur between 1690-1750^{-1}; the positive difference between the reactant C=O peak located at 1700 cm^{-1} (Fig.3B) and the product's peak at 1720 cm^{-1} (Fig.3D) would support succesful esterification. Thirdly, an intermediate-sized peak is observed both for the reactant (Fig.3C) and the product (Fig.3E) at 1630 and 1640 cm^{-1} respectively; their size and location are suggestive of nonconjugated C=C bond stretching (occuring between 1620-1680 cm^{-1}), which would be consistent with retention of the double bond during the proposed esterification. As an aside, clusters of peaks likely associated with C-H bond stretching were observed on both spectra (Fig.3F&3G) but inter-comparison was obfuscated by the strong and broad O-H stretching signal apparent in the reactant's spectrum.

Both ^{1}H and ^{13}C nuclear magnetic resonance spectroscopy was carried out on a sample of product using a "Bruker Avance III HD 500 MHz Smart Probe" spectrometer, with spectrograms produced at 500 MHz and and 125 MHz respectively. Chemical shifts are considered as downfield shifts (δ in ppm) from the tetramethyl silane standard ($\delta = 0$ ppm) and calibrated using the shift exhibited by trace amounts of non-deuterated chloroform ($\delta = 7.26$ ppm singlet for ^{1}H and $\delta = 77$ ppm triplet for ^{13}C) within the sample solvent. The chloroform-d solvent also introduced a trace amount of water which exhibited a singlet peak ($\delta = 1.63$ ppm) in the ^{1}H spectrogram; these peaks were ignored when examining the spectra for dimethyl itaconate, as presented here:

Dimethyl itaconate: ^{1}H NMR ($CDCl_3$, 500 MHz): δ 3.34 (d, 2H, $J = 0.71$ Hz), 3.69 (s, 3H), 3.76 (s, 3H), 5.71 (q, 1H, $J = 0.97$ Hz), 6.32 (s, 1H). ^{13}C {^{1}H} NMR ($CDCl_3$, 125 MHz): δ 37.5, 52.1, 52.2, 128.6, 133.7, 166.6, 171.1. [GO BACK OVER THIS WITH THE AID OF "AN OLD CAPTIVITY NOTES FROM 22ND JANUARY 2025"]

A quantitative ^{1}H spectrogram showed water constituting a 5 % w/w impurity, al-

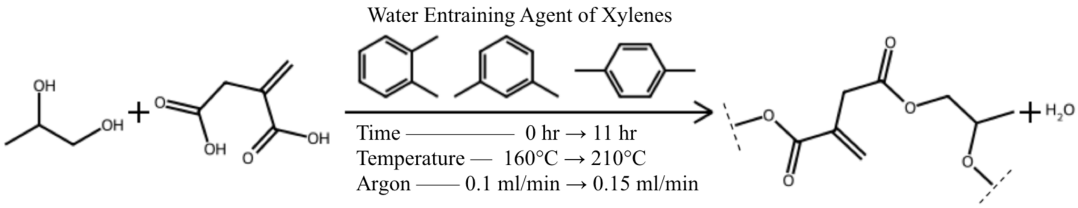

Figure 4: *Simple reaction scheme for synthesis of poly(propylene itaconate).*

though the dimethyl itaconate sample purity is expected to exceed 95 % given that the deuterated solvent will have introduced an unknown amount of water.

2.5 Synthesis of Unsaturated Polyester Resin

Unsaturated polyester was synthesised via the polyesterification of itaconic acid with propylene glycol to produce poly(propylene itaconate), as described by the simplified reaction scheme in Figure 4. This was achieved using the melt polycondensation method since a precedent exists in the literature for its succesful use in the synthesis of linear unsaturated polyesters derived from itaconic acid [18, 67, 54, 53]. The stoichiometric ratio between itaconic acid and propylene glycol was set at 1:1.1, on the basis that propylene glycol is lost during the heating process which reaches temperatures exceeding it's boiling point [55]. Figure 5 shows the apparatus used to synthesise unsaturated polyester; note, a Dean-Stark trap (Fig.5I) has been used allowing the 'azeotropic method' of polyester synthesis (as opposed to the 'dry' method) which reduces the time taken in synthesising a high molecular weight polyester [55]. In this method, an entraining agent (mixed xylenes) which can form an azeotrope with the water is added to the mixture (5 % w/w) [38, 8]; this increases the rate at which water is condensed and trapped by the Dean-Stark, driving the reaction forward by removing the unwanted byproduct of water from the reaction mixture. Furthermore, argon gas was used to sparge the reaction mixture (Fig.5E) facilitating the removal of water from the bulk mixture and mitigating the combustion risk associated with high temperature xylenes [55].

The procedure was initiated by charging the four-necked round-bottomed flask (Fig.5D) with the requisite masses of the two monomers and xylenes, opening the argon taps to provide a 5 bubbles per second of flow of sparging gas through the argon injector (Fig.5E) [55], opening the water taps to actively cool the Liebig condenser (Fig.5I), setting the overhead stirrer to a speed of 50 rpm (Fig.5G) [55], and bringing the reaction mixtures temperature (Fig.5F) up to 160 degreeCelsius using the hotplate (Fig.5A). During the procedure the mixture's temperature was raised monotonically by 10 $°\mathrm{C\,h^{-1}}$ until it reached a maximum of 210 °C, at which it was held.

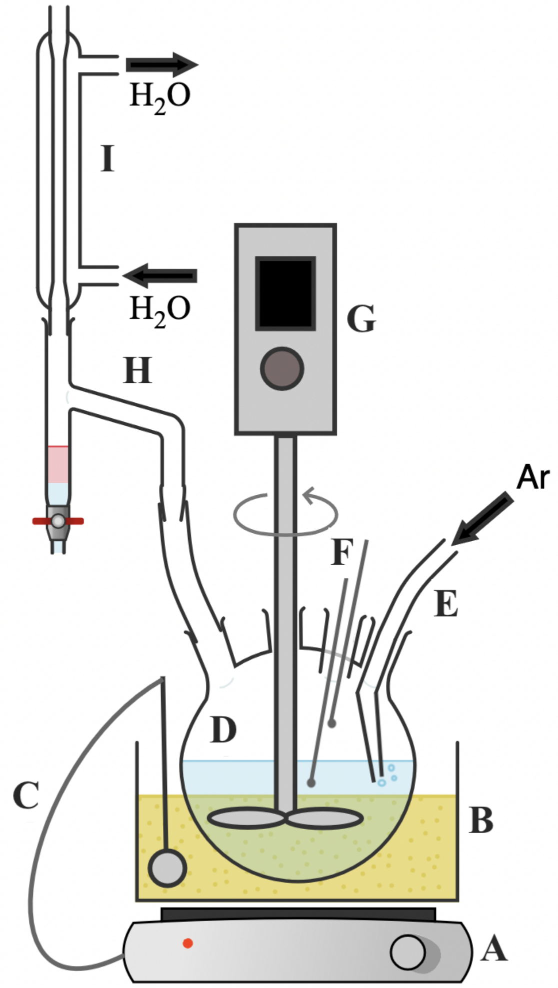

Figure 5: *Apparatus for synthesis of unsaturated poly(propyl itaconate) [10]. (A) Hotplate stirrer (B) Sand bath (C) Temperature controller probe (D) Four-necked round-bottomed flask containing the reaction mixture (E) Stainless-steel argon injector (F) Probes for batch and vapour tamperatures (G) Overhead stirrer (H) Dean-Stark trap (G) Liebig condenser.*

Reaction progress was ascertained by periodically retrieving a sample of the reaction mixture and ascertaining it's acid number, whose unit is $mg\,g^{-1}$ describing the milligrams of potassium hydroxide required to neutralise a gram of sampled unsaturated polyester. This rough form of end-group analysis is commonly used in unsaturated polyester synthesis [54, 67, 14, 55] and provides a proxy value estimating the number of un-esterified acid groups remaining within the unsaturated polyester. An American Society for Testing and Materials standard ("ASTM D4662-20") was used for this purpose. In-line with common practice, heating of the polyesterifying mixture was ceased at an acid value of 50 [55, 8, 38]. On cessation of heating, an outlet was opened for egress of remnant xylenes and the unsaturated polyester was allowed to cool in an air bath.

Since cooling unsaturated polyester will undergo a monotonic increase in viscosity until it reaches room temperature where it exists as a solid thermoplastic mass; it is therefore common to prepare the unsaturated polyester resin immediately after cessation of heating by adding reactive diluent directly to the cooling unsaturated polyester [55]. Unfortunately, reactive diluents, by their very nature, may be liable to polymerise during this addition causing a premature gelation (untimely and incomplete cross-linking of unsaturated polyester resin usually culminating in the genesis of a rubber-like solid). Mitigation against the dual-threat of hardening into thermoplastic and gelation into rubber is commonly enacted by awaiting a suitable intermediate temperature and addition of an addition-polymerisation inhibiting agent [55, 1, 13]. In this work, we awaited a temperature of 100 °C before adding the requisite masses of inhibitor and reactive diluent. This work adopted the commonly-used free-radical scavenger hydroquinone as an inhibiting agent, and this was added so as to make up 0.06 % w/w of the cooling unsaturated polyester mixture; hydroquinone was dissolved in a small amount of propylene glycol before addition to ensure homogeneity during it's mixing into the bulk [55]. The inhibitor percentage was determined through a process of trial-and-error, where lower percentages tended to facilitate premature gelation of unsaturated polyester resins on the shelf, whilst higher percentages caused problems during the curing procedures. Note, the percentage of inhibiting agent was above that which is commonly used in the synthesis of conventional unsaturated polyester resins (approximately 0.02 %) [55, 13]; this was the consequence of both the relatively higher proportion of unsaturated sites along the polyester backbone and a higher copolymerisation affinity between the reactive diluent and the sites of unsaturation. Subsequent of introduction and homogenisation of inhibiting agent into the unsaturated polyester mixture, the reactive diluent (dimethyl itaconate) was added such that it made up 30 % w/w of the final unsaturated polyester resin. Reactive diluent content was intentionally less than the commonly used optimal ratios seen in literature, as this was a characteristic that was to be optimised for [55, 1, 13, 8].

2.6 Unsaturated Polyester Resin Characterisation

[WRITE ABOUT GPC OR NMR IF COMPLETED]

2.7 Composite Preparation

Figure 6 presents an overview of the method by which cured thermoset composite samples of various compositions were prepared. Given the danger of self-accelerating decomposition reactions and associated thermal runaway, an external manufacturer was responsible for the mixing and provision of a safely-phlegmatised base initiating agent comprised of dibenzoyl peroxide and calcium sulphate (Fig.6F); this arrived pre-mixed in a set ratio of 2:8 between the dibenzoyl peroxide ($1 - c_2 = 0.2$ w/w) and the calcium sulphate ($c_2 = 0.8$ w/w). Base unsaturated polyester resin was mixed (Fig.6G) in a set ratio of 3:7 between unsaturated polyester resin ($c_1 = 0.7$ w/w) and dimethyl itaconate ($1 - c_1 = 0.3$ w/w) as described previously; the elected ratio represents the lowest amount of reactive diluent possible before viscosity precluded the possibility of further mixing- this increased the size of the possible parameter space to be interrogated (by increasing the range of modified unsaturated polyester resin reactive diluent concentrations that could be mixed to by means of dilution).

Modified unsaturated polyester resin was mixed (Fig.6E) in a varying ratio of $x_1 : (1-x_1)$ between the base unsaturated polyester resin (x_1 w/w) and the dimethyl itaconate ($1 - x_1$ w/w); by associating the x_1 parameter with a range ($[0.45, 1.00]$) within which it could vary, we fully describe the first predictor variable to be optimised (x_1, unsaturated polyester content modifier). Modified initiating agent was mixed (Fig.6D) in a varying ratio of $x_2 : (1-x_2)$ between the base initiating agent (x_2) and calcium sulphate $(1-x_2)$; x_2 was provided a range ($[0.4, 1.00]$) hence describing the second predictor variable to be optimised (x_2, dibenzoyl peroxide content modifier). Initiated unsaturated polyester resin was mixed (Fig.6C) in a varying ratio of $x_3 : (1-x_3)$ between modified unsaturated polyester resin (x_3) and modified initiating agent $(1 - x_3)$; x_3 was provided a range ($[0.5, 0.95]$) which therewith described the final predictor to be optimised (x_3, initiating agent content modifier).

Initiated unsaturated polyester resin samples were thermochemically cured into rigid thermoset plastic samples by exposing them to elevated temperatures in a drying oven (Fig.6B); the curing profile was fixed for all samples: 3 hours at 80 °C followed by 1 h at 120 °C. All cured thermoset samples were machined to the same dimensions (Fig.6A) in preparation for characterisation; this was achieved by a grinding process.

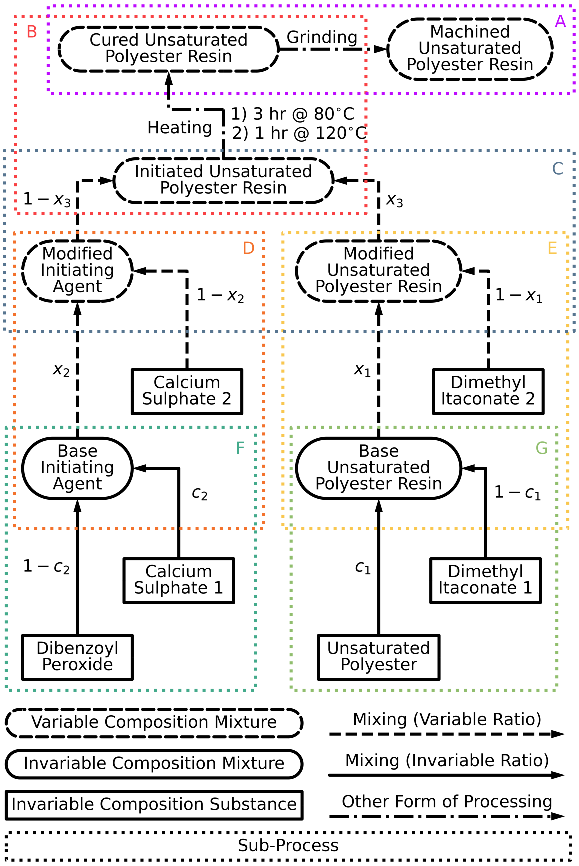

Figure 6: *Mixing and processing schedule for the manufacture of thermoset composite samples. (A) Machining. (B) Thermochemical curing. (C) Mixing for initiated sample. (D) Mixing for modified initiator. (E) Mixing for modified resin. (F) Manufacturer's mixing for base initiator. (G) In-house mixing for base resin.*

2.8 Composite Characterisation

Given that a key aspect of a thermoset's utility lies in it's strength and stiffness within the elastic region of deformation we considered the yield strength and Young's modulus as legitimate objective variables of interest. Compressive testing was elected as a means by which these metrics could be obtained from samples based on both the simplicity with which a curing regime could be devised for the manufacture of samples of the requisite dimensions and the sparing mass of material required for these. An American Society for Testing and Materials standard was employed ("ASTM D695-23") in the devising of method; with the cylindrical test sample form, sample diameter of 12.7 mm, and the compression speed of 1.3 mm min^{-1} adopted. However, sample length was adapted down by half (to 12.7 mm) to reduce the mass of material required per sample.

Equipment Compressive testing was carried out using a "Tinius Olsen 50ST" universal test machine (50 kN capacity) fitted with compression platens. Standard software ("Horizon") purveyed by the machine's manufacturer was used in the control of the machine, retrieval of data from the load cells, and export of formatted data for further analysis. Exported data consisted of time ($\pm$ 0.16 µs), force ($\pm$ 1 mN), and platen displacement ($\pm$ 1 nm).

Euler Buckling To mitigate against concerns of Euler buckling taking place in samples of the proposed dimensions, we calculated the critical load based on Euler's critical load equation and some estimated properties for conventional polyester thermosets [24]. Equation 2 shows how the critical load (P_{cr} in N) is related to the elastic modulus of the material (E in Pa), the second moment of area (I in m^4), the effective length factor (K, unitless), and the initial length of the column in question (L in m).

$$P_{cr} = \frac{\pi EI}{(KL)^2} \tag{2}$$

Young's moduli for conventional thermoset polyesters were expected to range from 2-4.5 GPa [39]. The second moment of the area for the top surface of the cylindrical sample with radius 6.35 mm was calculated to be 1.276×10^{-9} m^4 by assuming it to be a perfect filled circular area, for which the generic equation 3 applied (where the r equated to the radius of the circle in m).

$$I_y = \frac{\pi}{4} r^4 \tag{3}$$

By assuming that both ends of the column would be braced against rotation or translation (given their compression between two platens), the theoretical column effective length factor would have been 0.5; as a further precaution a conservative design value of 0.65 was assumed based on pertinent literature [68]. On these assumptions it was estimated that the samples in question would be likely to Euler buckle between 433-976 kN. As the compressive strength expected of conventional unsaturated polyester thermosets were to be expected to fall within the range of 89-252 MPa and this corresponded to maximum compressive forces of between 11-32 kN (given the surface area of the samples), we were reassured that the samples were unlikely to experience Euler buckling.

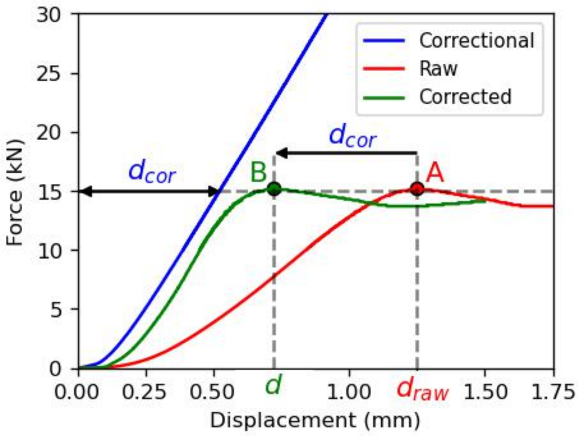

Figure 7: *Strain correction of an archetypical raw dataset; the legend notes the various datasets considered.*

Correction of Raw Data Imperfect rigidity of the test machine itself confounded the raw exported datasets generated for samples tested. By carrying out a compressive test with no sample between the platens, a force-displacement curve representing the inherent stiffness of the machine was generated- this was the correctional dataset. Equation 4 shows how the corrected displacement value (d in mm) could be obtained by taking a raw displacement value (d_{raw} in mm) and subtracting the corresponding correctional displacement value (d_{cor}).

$$d = d_{raw} - d_{cor} \tag{4}$$

Figure 7 presents the correctional dataset, an archetypical raw dataset, and it's corresponding corrected dataset. Point 'A' is corrected to produce point 'B' by using it's associated force to find d_{cor}, after which this can be subtracted from it's d_{raw} value to produce the d value that point 'b' obtains.

Processing of Corrected Data Force values from a sample's corrected dataset were processed into engineering stress values by taking into account the initial surface area. Each cylindrical sample's diameter was measured using a "Magnusson" digital caliper (accuracy ± 0.02 mm) from three different positions ($\o_1, \o_2, \o_3$ in mm), each offset by roughly +45° around the circumference, and the mean of these values ($\o_{avg}$ in mm) calculated (see Equation 5).

$$\o_{avg} = \frac{\o_1 + \o_2 + \o_3}{3} \tag{5}$$

Equation 6 shows how the initial top-face surface area was calculated from the average top surface diameter $\o_{avg}$.

$$A_{init} = 0.25 * \pi * \o_{avg}^2 \tag{6}$$

Subsequently, engineering stress (σ in MPa) could be calculated (Equation 7) by considering the force applied (F in N) and the initial top surface area, A_{init} [47].

$$\sigma_{eng} = \frac{F}{A_{init}} \tag{7}$$

Displacement values from a sample's corrected dataset were processed into engineering strain values by taking into account the initial length of the samples and the length of the sample during compression. Initial length (l_{init} in mm) was calculated (8) by taking the mean of three length measurements (l_1, l_2, l_3 in mm) (these taken in the same fashion as for diameter).

$$l_{init} = \frac{l_1 + l_2 + l_3}{3} \tag{8}$$

Equation 9 shows the simple link between length of a sample at a given point during a compressive test (l in mm), the initial sample length (l_{init}) and the corrected displacement at that point d.

$$l = l_{init} - d \tag{9}$$

Subsequently, engineering strain was calculated (Equation 10) by finding the absolute change in sample length (Δl_{init} in mm) and dividing this by the initial length of the sample (l_{init}).

$$\epsilon_{eng} = \frac{\Delta l_{init}}{l_{init}} = \frac{l - l_{init}}{l_{init}} \tag{10}$$

Stress-Strain Smoothing Sample-rate reduction and filtering were applied simultaneously during the downsampling of the stress-strain dataset resulting in a smoothed dataset. Downsampling was achieved using a windowed averaging technique, whereby all the points within the unsmoothed dataset as a whole (p_{us}) was split (Eq.11) into a number of windows along the strain-axis (p_s) by electing a suitable number of points per window (p_w). Suitability was determined by trial-and-error within the wider experimental framework but was also based on the requirement of a perfect divisibility of p_{us} by p_w. Given the constant size of the unsmoothed dataset ($p_{us} = 1,000$) a constant number of points per window was elected ($p_w = 5$).

$$p_s = p_{us}/p_w \tag{11}$$

Equation 12 shows how the processs of averaging stress (σ^i_{eng}) and strain (ϵ^i_{eng}) values within a given window yielded a single point described by it's average stress ($\overline{\sigma}^j_{eng}$) and strain ($\overline{\epsilon}^j_{eng}$).

$$\overline{\sigma}^j_{eng}, \overline{\epsilon}^j_{eng} = \frac{\sum_{i=1}^{p_w} \sigma^i_{eng}}{p_w}, \frac{\sum_{i=1}^{p_w} \epsilon^i_{eng}}{p_w} \tag{12}$$

Averaged points from each window were assembled to form the new downsampled dataset depicted in Equation 13.

$$[[\overline{\sigma}^j_{eng}, \overline{\epsilon}^j_{eng}], [\overline{\sigma}^{j+1}_{eng}, \overline{\epsilon}^{j+1}_{eng}], ..., [\overline{\sigma}^{p_s}_{eng}, \overline{\epsilon}^{p_s}_{eng}]] \tag{13}$$

Figure 8 demonstrates the efficacy of this technique in removing noise from the unsmoothed stress-strain dataset.

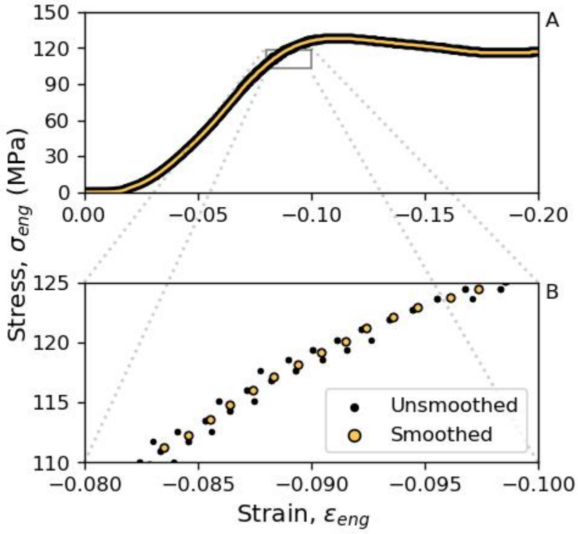

Figure 8: *Multi-scale depiction of the stress-strain dataset before and after downscaling. (A) Overview line plot. (B) Zoomed scatter plot.*

Derivative Analysis The proportional limit may be defined as the stress at which a material fails to abide by Hooke's law (e.g. stress fails to proportionally rise with strain)- the gradient of the preceding elastic region of deformation being equal to the Young's modulus. Given the material's unreliability in producing stress-strain curves with an obvious linear region, the limit of proportionality was equated to the point at which the stress-strain curve began to level-off (i.e. the first point of inflection). To find this inflection point, the first derivative of stress with respect to strain (tangent modulus E_t in MPa) was calculated from the smoothed stress-strain dataset (Eq.14) by taking into account the instantaneous change in both stress ($\Delta\sigma_{eng}$) and strain ($\Delta\epsilon_{eng}$).

$$E_t = \frac{\Delta\sigma_{eng}}{\Delta\epsilon_{eng}} = \frac{\sigma^{j+1}_{eng} - \sigma^j_{eng}}{\epsilon^{j+1}_{eng} - \epsilon^j_{eng}} \tag{14}$$

Figure 9 shows how smoothing of the stress-strain dataset facilitated the analysis of the resultant tangent modulus versus strain dataset. Note that markedly step-like behaviour in the raw stress-strain data (Fig.9) was an artifact of an unknown processing step applied by the universal test machine's proprietary software.

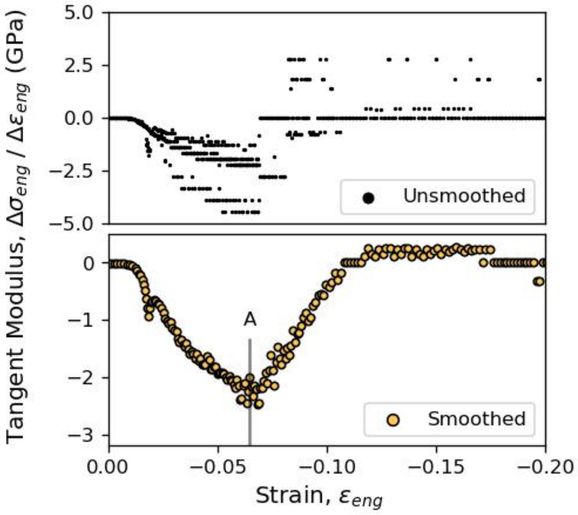

Figure 9: *A comparison of the tangent modulus vs strain curves obtained from both the raw and the smoothed counterpart stress-strain datasets. (A) Strain at first inflection point.*

Detection of the first inflection point was automated using a simple algorithm that iterated chronologically through the tangent modulus versus strain dataset from low strains to high. At each point the algorithm would calculate an average gradient exhibited by an equal number of points both ahead and behind (the 'point horizon') the current strain value; cessation and return of the current strain was enacted on reaching a point at which summed gradient ahead and behind fell below a threshold of zero. Figure 9A shows how a point horizon of 10 was used to identify the first inflection point, and therewith the strain at which the limit of proportionality had been reached.

Objective Retrieval By re-considering the smoothed stress-strain dataset, the point with the closest proximity to the proportional limit's determined strain value was equated to the first inflection point: the proportional limit (Fig.10A).

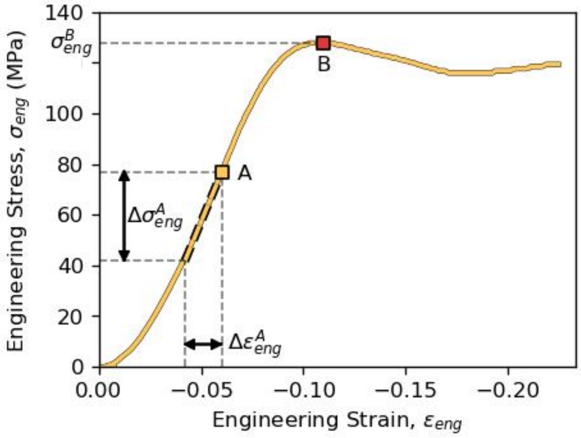

Figure 10: *Means by which Young's modulus and yield point were acquired from an archetypical stress-strain dataset. (A) First inflection point. (B) Second inflection point.*

As shown in Figure 10, a polynomial was fit to the ten points preceding the proportional limit; allowing approximation of Young's modulus (E in MPa) from the change in stress (Fig.10: $\Delta\sigma^A_{eng}$) and strain (Fig10: $\Delta\epsilon^A_{eng}$) in accordance with Equation 15 [34]. Outputted Young's moduli were negative given the compresive nature of the dataset, but were naturally taken as absolute values.

$$E = \frac{\sigma_{eng}}{\epsilon_{eng}} = \frac{\Delta\sigma^A_{eng}}{\Delta\epsilon^A_{eng}} \tag{15}$$

Based on it's common definition ("the stress at which a material begins to behave in a plastic fashion"), this work defined (Eq.16) the yield strength (σ_y in MPa) as the stress required to reach the second inflection point (Fig.10B) as visualised by σ^B_{eng} in Figure 10 [76].

$$\sigma_y = \sigma^B_{eng} \tag{16}$$

Since the samples tested consistently produced similar stress-strain curves to that depicted in Figure 10, the second inflection point was trivially located searching the stress-strain dataset for the point exhibiting the highest stress value; the sample's yield or break strength.

2.9 Optimisation

As described above, three predictors controlled the composition of samples: the unsaturated polyester content ratio (x_1), the dibenzoyl peroxide content modifier (x_2) and the initiating agent content modifier (x_3). These three predictors and their ranges formed a d-dimensional ($d = 3$) parameter space ($\mathcal{X}$) which is a subset of the entire real coordinate space ($\mathcal{X} \subseteq \mathbb{R}^d$). This parameter space was explored by the sequential optimisation techniques, with the objective variables including Young's modulus (E) and yield strength (σ_y).

Three different optimisation procedures were carried out from the same initial prior dataset: two were single-objective Bayesian routines with differing objectives, whilst the third was a purely exploratory method which was objective-indifferent. A fourth multi-objective Bayesian optimisation routine incorporating both the objectives previously considered was carried out using a combination of the original prior dataset and all data generated by the three sequential optimisations.

Single-Objective Experiments Optimisation problems were based on black-box functions f which took any point within the parameter space $\mathcal{X}$ and returned a real-valued target $\mathbb{R}$; this being a mapping between the two respective spaces ($f : \mathcal{X} \to \mathbb{R}$). Equation 17 describes optimisation as the maximisation of the black-box function by tuning the predictor variable values ($\mathbf{x} = [x_1, x_2, x_3]$) such that the global optimum set of predictors ($\mathbf{x}^* = [x_1^*, x_2^*, x_3^*]$) is sought.

$$\mathbf{x}^* = \underset{\mathbf{x}\in\mathcal{X}}{\arg\max}\, f([x_1, x_2, x_3]) \tag{17}$$

Equation 18 describes the two black-box functions which were separately maximised during the course of the single-objective optimisations; the former f_E and latter f_σ considering Young's modulus E and yield strength σ_y as target variables respectively.

$$\begin{aligned} E &= f_E(\mathbf{x}) \\ \sigma_y &= f_\sigma(\mathbf{x}) \end{aligned} \tag{18}$$

Eight vectors of predictor variables, each corresponding to a sample (i), were representatively drawn by Sobol sampling from across the parameter space $\mathcal{X}$ forming an initial matrix of predictors ($\mathbf{X}$); the matrix's form is presented in Equation 19.

$$\begin{aligned} \mathbf{X} =&[\mathbf{x}^{i=1}, \mathbf{x}^{i=2}, ..., \mathbf{x}^{i=8}] \\ =&[[x_1^{i=1}, x_2^{i=1}, x_3^{i=1}], [x_1^{i=2}, ...], ...] \end{aligned} \tag{19}$$

Evaluation of all the vectors of the predictor matrix ($f_E(\mathbf{X})$ or $f_\sigma(\mathbf{X})$) generated a vector of targets ($\mathbf{E}$ or $\boldsymbol{\sigma_y}$). Combination of the initial predictor matrix $\mathbf{X}$ and the corresponding targets $f(\mathbf{X})$ produced an initial dataset ($\mathcal{D} = [\mathbf{X}, f(\mathbf{X})]$) with which sequential optimisation experiments were begun. During the course of a sequential optimisation experiment, 16 further samples were drawn from across the parameter space $\mathcal{X}$, with one sample taken per iteration; each sample's predictor vector $\mathbf{x}$ and function evaluation $f(\mathbf{x})$ was appended to the growing dataset $\mathcal{D}$ associated with that experiment. At each iteration the Bayesian experiments strove to balance exploration and exploitation during selection of sample composition within the parameter space, whilst the benchmark deterministic experiment solely focussed upon exploration.

Bayesian optimisation involved an iterative process of developing a surrogate model approximating the latent objective function based on the available dataset $\mathcal{D}$, using an acquisition function to evaluate candidate suitability and selection of a new sample $\mathbf{x}^i$, and finally evaluation of a sample followed by incorporation of the resultant target scalar σ_y^i and predictor vector $\mathbf{x}^i$ into the growing dataset $\mathcal{D}$.

On updating the dataset $\mathcal{D}$, Gaussian process regression was used to develop a Gaussian process regression model $\mathcal{GP}$ which can be considered a distribution over possible functions ($f(\mathbf{x}) \sim \mathcal{GP}(m(\mathbf{x}), K(\mathbf{x}, \mathbf{x}'))$) fully described by both it's mean (m) and covariance (K) functions. Computationally, the Gaussian process was modelled by a high-dimensional multivariate Gaussian distribution $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ described by it's mean vector $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$; each succesive iteration saw the prior process conditioned on new data producing an updated posterior process.

For the purpose of encoding our prior beliefs in process smoothness we employed the Matérn family of covariance functions; specifically, one employing $\nu = 5/2$ which yields Equation 20 based on Euclidean distance (d). The $\nu = 5/2$ case balanced the need for smoothness in sampled paths with the drawbacks of higher values of ν; as $\nu \to \infty$, the sampled paths become infinitely differentiable, making them unrealistically smooth for practical application [65] and potentially interfering with succesful calculation of the likelihood [69].

$$K_{M5/2}(x, x') = (1 + \sqrt{5}d + \frac{5}{3}d^2)exp(-\sqrt{5}d) \\ \text{where } d = |x - x'| \tag{20}$$

Each iteration saw the use of a one-step lookahead sampling policy utilising a simple reward utility function (Equation 21) which takes the current dataset $\mathcal{D}$ and of all samples $\mathbf{X}$ returns the sample $\mathbf{x}$ exhibiting the highest target (ϕ; in practice either E or σ_y) according to the posterior process's mean function $max\ m(X)$.

$$u(\mathcal{D}) = max\ m(\mathbf{X}) \tag{21}$$

Utility quantification of a putative sample $\mathbf{x}_p$ was calculated by the "expected improvement" acquisition function α_{EI} (Equation 22) which calculates the marginal gain in utility ($u(\mathcal{D}') - u(\mathcal{D})$) between the incumbent best target value in the current dataset (ϕ^*) and that of a new dataset containing the putative $\mathbf{x}_p$ value ($u(\mathcal{D}')$). The second line of Equation 22 clarifies how the calculation is one for the expected marginal gain, whilst the third presents reformulation for the noiseless case; this equation exemplifies how putative points are rewarded based on both scale and probability of improvement effectively trading off exploitation and exploration. This work practically applied a more sophisticated logged noise-aware implementation ("qLogNoisyExpectedImprovement" or "qLNEI") which has recently seen successful use in high-dimensional computational domains [3].

$$\begin{aligned} &\alpha_{EI}(\mathbf{x}_p, \mathcal{D}) = u(\mathcal{D}') - u(\mathcal{D}) \\ &= \mathbb{E}[max\, m_{\mathcal{D}'}(\mathbf{X}')] - max\, m_{\mathcal{D}}(\mathbf{X}) \\ &= \int_{\phi^*}^{\infty} \phi \mathcal{N}(\mu, \sigma^2) d\phi - \phi^* \int_{\phi^*}^{\infty} \mathcal{N}(\mu, \sigma^2) d\phi \end{aligned} \tag{22}$$

Trivial optimisation of the acquisition function (Equation 23) was achieved by generating a large set of Sobol sampled points across the parameter space $\mathbf{X}_p$, after which cheap evaluation facilitated identification of the candidate sample with the maximal utility score $\mathbf{x}^+$.

$$\mathbf{x}^+ = \underset{\mathbf{X}_p \in \mathcal{X}}{\arg\max}\, \alpha(\mathbf{X}_p) \tag{23}$$

The objective-indifferent space-filling benchmark sampling exercise was based on a sequential Monte Carlo strategy by the name of "mc-intersite-proj-th" ("MIPT") [17]. Utility was quantified by the conditional acquisition function displayed in Equation 24, which took both a putative point $\mathbf{x}_p$ and the current dataset's previous sample coordinates $\mathbf{X}$ and returns either zero or the Euclidean distance to the nearest previously sampled point k depending on whether projected distance is below or in excess of the d_m value respectively. The d_m is dependent upon the number of previous samples n and the arbitrary constant s which has been found to be optimally set to 0.5 for a reasonable trade-off between intersite distance and projected distance.

$$\begin{aligned} &\alpha_{mipt}(\mathbf{x}_p, \mathbf{X}) = \\ &\begin{cases} 0 & if \min_{\forall \mathbf{x}_i \in X} ||\mathbf{x_i} - \mathbf{x_p}||_{-\infty} < d_m \\ k, & if \min_{\forall \mathbf{x}_i \in X} ||\mathbf{x_i} - \mathbf{x_p}||_{-\infty} \geq d_m \end{cases} \\ &\text{where } d_m = \frac{2s}{n} \text{ and } k = \min_{\forall \mathbf{x}_i \in X} ||\mathbf{x_i} - \mathbf{x_p}||_2 \end{aligned} \tag{24}$$

Multi-Objective Experiment Given the desirability of both Young's modulus and yield strength in a thermoset composite, a dual-objective Bayesian optimisation experiment was attempted, with the *a posteriori* goal of sample-efficiently identifying the pareto front ($\mathcal{P}$) [28]. The Pareto front consists of all samples which remain unsurpassed (by any other sample evaluated) in terms of at least one of their target variables (E or σ_y). This Pareto front allows a rational aagent to make subjective yet informed decisions about material composition in light of expected trade-offs in end properties. Equation 25 frames the multi-objective optimisation problem in terms of a dual-objective function $\mathbf{f}$ (comprised of the single-objective functions f_E and f_σ) which returns a vector $\mathbf{y}$ of the two targets (E and σ_y).

$$\begin{aligned} \mathbf{y} &= \mathbf{f}(\mathbf{x}) \\ \{E, \sigma_y\} &= \{f_E(\mathbf{x}), f_\sigma(\mathbf{x})\} \end{aligned} \tag{25}$$

An observed Pareto front may be obtained directly from a given dataset $\mathcal{D}$ using the function $f_\mathcal{P}$ (Eq. 26).

$$\begin{aligned} \mathcal{P} &= f_\mathcal{P}(\mathcal{D}) \\ &= \{\mathbf{y} | \mathbf{y} \in \mathbf{Y}, \nexists\, \mathbf{y}' \in \mathbf{Y} s.t.\, \mathbf{y}' \succ \mathbf{y}\} \end{aligned} \tag{26}$$

Alternatively, where a multi-objective function $\mathbf{f}$ is considered alongside a matrix of predictor variables $\mathbf{X}$, the function $f^*_\mathcal{P}$ may be used to produce a Pareto front (Eq. 27).

$$\begin{aligned} \mathcal{P} &= f^*_\mathcal{P}(\mathbf{X}; \mathbf{f}) \\ &= \{\mathbf{f}(\mathbf{x}) | \mathbf{x} \in \mathbf{X}, \mathbf{f}(\mathbf{x}) \succ \mathbf{f}(\mathbf{x}') \forall \mathbf{x}' \in \mathbf{X}\} \end{aligned} \tag{27}$$

By retrieving all the raw mechanical test data associated with the samples evaluated in the three sequential experiments and their common prior, a 56-sample dataset $\mathcal{D}$ was generated comprised of a matrix of sampled predictors $\mathbf{X}$ and a matrix of their respective targets $\mathbf{Y}$ ($\mathbf{E}$ and $\boldsymbol{\sigma_y}$). Having conditioned two separate Gaussian processes on $\mathbf{E}$ and $\boldsymbol{\sigma_y}$ we obtained two separate posterior processes ($\mathcal{GP}_E$ and $\mathcal{GP}_\sigma$); these provided initial prior processes for all subsequent sequential optimisation. The pair of resulting processes were respectively described mathematically by their mean (m_E, m_σ) and covariance functions (K_E, K_σ) and computationally by their multivariate normal analogues ($\boldsymbol{\mu}_E$, $\boldsymbol{\mu}_\sigma$ and $\boldsymbol{\Sigma}_E$, $\boldsymbol{\Sigma}_\sigma$). Equation 28 shows how a dual-objective surrogate function $\tilde{\mathbf{f}}$ may be sampled from the pair of Gaussian processes ($\mathcal{GP}_E$ and $\mathcal{GP}_\sigma$) previously conditioned.

$$\begin{aligned} \tilde{\mathbf{f}} &\sim \{\mathcal{GP}_E, \mathcal{GP}_\sigma\} \\ \{\tilde{f_E}, \tilde{f_\sigma}\} &\sim \{\mathcal{GP}(m_E, K_E), \mathcal{GP}(m_\sigma, K_\sigma)\} \end{aligned} \tag{28}$$

Utility of a given Pareto front was evaluated using the hypervolume indicator utility function u_{hv} presented in Equation 29, which is equal to the d-dimensional Lebesgue measure λ_d of the space dominated by a Pareto front $\mathcal{P}$ and bounded from below by reference point r (r and v describe vertices).

$$u_{hv}(\mathcal{P}; r) = \lambda_d(\bigcup_{v \in \mathcal{P}} [r, v]) \tag{29}$$

Equation 30 shows how the utility function u_{hv} is used in the subsequent calculation of the one-step marginal gain known as the "hypervolume improvement" between the extant $\mathcal{P}$ and the updated Pareto front $\mathcal{P}'$ (which includes the putative point $\mathbf{x}_p$).

$$f_{hvi}(\mathcal{P}, \mathcal{P}'; r) = u_{hv}(\mathcal{P}'; r) - u_{hv}(\mathcal{P}; r) \tag{30}$$

The common multi-objective acquisition function by the name of "expected hypervolume improvement" is described by Equation 31, which takes a putative point $\mathbf{x}_p$ and the current dataset $\mathcal{D}$ and returns the instantaneous improvement metric. Expectation manifests itself during the speculative update of the dataset $\mathcal{D}$ to $\mathcal{D}'$, where the two targets are drawn from their respective marginal Gaussian distributions (of their respectively updated posterior Gaussian processes) located at the putative point $\mathbf{x}_p$; whose expectation ($\mathbb{E}(\mathbf{x}_p)$) is naturally the mean of the marginal at that location ($m_E(\mathbf{x}_p)$ or $m_\sigma(\mathbf{x}_p)$).

$$\begin{aligned} &\alpha_{ehvi}(\mathbf{x}_p; \mathcal{D}) = f_{hvi}(f_\mathcal{P}(\mathcal{D}'), f_\mathcal{P}(\mathcal{D}); r) \\ &\text{where } \mathcal{D}' = \mathcal{D} \cup \{\mathbf{x}_p, m_E(\mathbf{x}_p), m_\sigma(\mathbf{x}_p)\} \end{aligned} \tag{31}$$

This work adopted Daulton's reformulation of the classic acquisition function (Eq. 31) which takes into account the noisy nature of objective function evaluations [21]; although the true formulation contains an integral which is analytically intractable, it may be approximated using Monte Carlo integration. Equation 32 describes the integral-aprroximating reformulation of the acquisition function $\hat{\alpha}$ which takes a putative point $\mathbf{x}_p$ and the dataset $\mathcal{D}$ and returns an instantaneous improvement value. This "noisy expected hypervolume improvement" acquisition function iterates from $1, \ldots, N$ using the t counter, with each iteration drawing a random dual-objective function $\tilde{\mathbf{f}}_t$ from the posterior gaussian process (see Eq. 28). This function $\tilde{\mathbf{f}}_t$ is used to speculatively produce Pareto fronts ($\mathcal{P}$ and $\mathcal{P}'$) for both an original dataset $\mathbf{X}$ and the updated dataset $\mathbf{X} \cup \mathbf{x}_p$ respectively, before the hypervolume improvement is gauged using f_{hvi} and the many trials N averaged.

$$\hat{\alpha}_{nehvi}(\mathbf{x}_p; \mathbf{X}) = \frac{1}{N} \sum_{t=1}^{N} f_{hvi}(f^*_{\mathcal{P}}(\mathbf{X} \cup \mathbf{x}_p; \tilde{\mathbf{f}}_t), f^*_{\mathcal{P}}(\mathbf{X}; \tilde{\mathbf{f}}_t); r) \tag{32}$$

This work practically applied the log implementation of the noise-aware α_{nehvi} acquisition function "qLogNoisyExpectedHypervolumeImprovement" ("qLNEHVI") which was shown to improve Bayesian optimisation performance in noisy settings [21].

Implementation All experiments were implemented in the Python programming language, with functionality spanning initial sampling techniques, adaptive sequential sampling methods, and an experimental workflow management system facilitating human-in-the-loop laboratory work; this package had the acronym "BO4ACST" (version 1.0.0)- "Bayesian optimisation for atmospheric carbon sequestering thermosets" [33]. Note that Bayesian sequential optimisation techniques including Gaussian process regression, gaussian process conditioning, and sample utility evaluation were facilitated by the open-source "adaptive experimentation platform" or "Ax" (version 1.1.2) [25]. Note that "Ax" makes use of a great deal of code from "Botorch", especially in it's implementation of custom Bayesian optimisation routines (e.g. non-standard acquisition function implementations) [58].

3 Results

All results stem from the datasets generated during the practical-application of Bayesian optimisation in the domain of thermoset composite property development. Practical experimental results were split into those derived from the two single-objective experiments and the singular multi-objective optimisation. Computational results centre on the re-analysis of datasets to ascertain the efficacy of the techniques applied.

3.1 Single-Objective Experiments

Differing between the two single-objective Bayesian optimisation experiments devised was the objective variable considered by the Bayesian sequential sampler, which was either Young's modulus E or yield strength σ_y. Initial Gaussian process priors developed for both experiments were trained on the same eight quasirandomly sampled (Sobol) points $\mathbf{X}$ (from across the parameter space $\mathcal{X}$) and their evaluations $\mathbf{Y}$. Both experiments proceeded sequentially with a single sample evaluated at each Bayesian iteration, the sample limit was set to 24 (including the prior-forming dataset), and the optimisation was one of objective metric maximisation.

An objective-naive sequential sampling technique ("MIPT") was used to sample in a purely exploratory fashion, which provided a benchmark against which Bayesian optimisation routine progression was gauged. Given it's objective-naivety, purely exploratory nature, and the fact that it was initiated on the same prior-forming dataset as the Bayesian routines, the MIPT routine was only run once (samples having both Young's modulus and yield strength evaluated); the resultant dataset could be directly compared with both the Bayesian sequential routines.

3.1.1 Young's Modulus Experiment

Figure 11 compares the optimisation progression of both Bayesian and exploratory sequential routines in the maximisation of Young's modulus. The one-shot quasirandomly-sampled initial dataset provided a baseline maximum of 4274 MPa and a mean of 3052 MPa. On reaching the sample limit, the Bayesian technique had exceeded both these baselines (max 5012 MPa, mean 3396 MPa) whilst the exploratory sampler only exceeded the maximum baseline (max 4851 MPa, mean 3046 MPa). Although the Bayesian routine ultimately discovered a sample exhibiting the highest Young's modulus within this experiment, it is interesting that the exploratory routine discovered a greater number of points surpassing the baseline maximum set by the initial dataset. The exploratory sampling routine was the first to locate a sample improving upon the baseline maximum; this was it's 15th sample with a Young's modulus of 4851 MPa. Subsequently, on it's 17th sample, the Bayesian routine discovered the experiments highest Young's modulus of 5012 MPa.

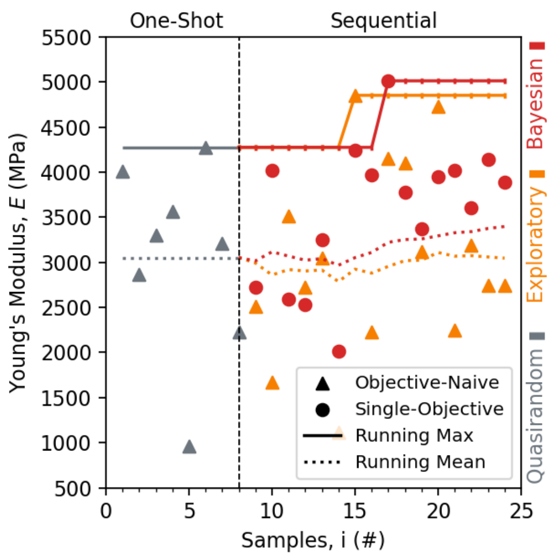

Figure 11: *Progression of a sequential optimisation experiment concerning the maximisation of Young's modulus.*

It is notable that this represented 69% the Young's modulus exhibited by this materials conventional counterpart (styrene-based unsaturated polyester resin) optimised under the exact same conditions (see 16).

3.1.2 Yield Point Experiment

Figure 12 compares optimisation progression of both the Bayesian and exploratory sequential routines with respect to the maximisation of the yield point. The one-shot quasirandomly-sampled initial dataset provided a baseline maximum of 137 MPa and a mean of 123 MPa. On reaching the sample limit, the Bayesian technique had surpassed the baseline maximum (143 MPa) and remained level with respect to the mean (123 MPa), whilst the exploratory routine had surpassed the baseline maximum (140 MPa) and failed to improve upon the baseline mean (118 MPa). In this experiment both the Bayesian and exploratory routines discovered the same number of points surpassing the baseline maximum, although the Bayesian technique ultimately uncovered the sample with the highest yield strength. The exploratory sampling technique was the first to discover a point improving upon the maximum baseline; with this happening on it's 15th sample with a yield strength of 140 MPa. Subsequently, the Bayesian sequential technique improves upon the maximum baseline on it's 18th sample (140 MPa) and surpasses the exploratory routine on the 21st sample with a final yield strength of 143 MPa.

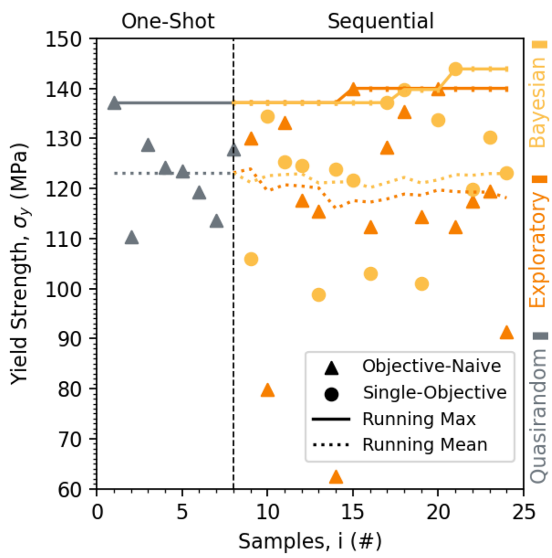

Figure 12: *Progression of a sequential optimisation experiment concerning the maximisation of yield strength.*

3.1.3 Cross-Comparison

In both single-objective optimisation experiments the Bayesian sequential routines out-performed the exploratory sampling routine in terms of discovery of samples approaching a global maximum within the alotted sample limit. Bayesian techniques draw benefit from adaptive sampling based on the balancing of exploitative and exploratory behaviour, the former exhibited clearly in the upward/constant trends shown by the Bayesian running means seen in both Figures 11&12. These results would support the hypothesis that the Bayesian sampling routine was more effective in maximising the objective property during the course of a sequential optimisation, but would need supporting with repeat experiments of either a physical or computational nature given the slim margins of improvement over the explorational technique in both experiments.

In both experiments we saw the exploratory sampling routine outperform the Bayesian technique in terms of sample efficiency: the exploratory technique improved upon the baseline maxima ($E = 4274$ MPa and $\sigma_y = 137$ MPa) of both experiments on drawing it's 15^{th} sample for evaluation ($E = 4851$ and $\sigma_y = 140$). The fact that this occurred on the 15^{th} sample for both experiments is trivially explained since this was the same exploratory routine being performed (i.e. the same samples being evaluated upon different target variables) and the especially well-performing 15^{th} sample was simply displaying it's evaluated properties in both experimental progressions; these circumstances almost come to pass a second time with respect to the 20^{th} sample drawn by the exploratory sampler. A hypothesis that the exploratory sampler more efficiently improves upon the baseline maximum than the Bayesian routine would be weakly supported by the above findings given that both circumstances were precipitated from the same data point, but further repeat studies in the physical or computational domain would aid in the supporting or negating this conjecture.

3.2 Multi-Objective Experiment

In the multi-objective setting a Bayesian sequential routine was performed which made use of the noisy expected hypervolume improvement acquisition function to sample-efficiently identify the Pareto front within an objective space comprised of both Young's modulus and yield strength. This dual-objective optimisation was facilitated by two separate Gaussian processes modelling the two different objective variables; both were trained on 56 samples concatenated from the re-analysed raw data of the previous single-objective experiments. Order of concatenation (and therewith re-indexing of samples numbers) was as follows: one-shot objective-naive quasirandom sampler (samples 1-8), sequential exploratory objective-naive routine (9-24), sequential Bayesian yield strength maximising routine (25-40), and the sequential Bayesian Young's modulus maximising routine (41-56). This prior-forming dataset $\mathcal{D}$ yielded an initial Pareto front $\mathcal{P}$ comprised of three samples: two having been discovered as maxima by Bayesian sequential routines in their two respective single-objective experiments with yield strength (sample 37) an Young's modulus (sample 49) as their objectives. whilst the third (sample 15) was derived from the exploratory technique as a point which performed well in both objectives of interest. With an ultimate sample limit of 72 and a single sample drawn per Bayesian iteration, the sequential multi-objective Bayesian routine proceeded to develop the Pareto front over 15 further samples (57-71).

All samples both prior-forming and sequentially-obtained are depicted in Figure 13, with the initial Pareto front marked as a dashed black line (Fig.13A). Given the initial Pareto front and considering only the direct evaluations obtained for samples it can be said that five samples were discovered which improved upon the baseline described by the Pareto front (samples 62,66,68,69,71).

Under progression of the multi-objective routine, the first novel pareto optimal point which improved the Pareto front was the 62$^{\text{nd}}$ sample which dominated the 37$^{\text{th}}$ sample shifting the boundaries of the developing Pareto front (Fig.13B). Subsequently, the 66$^{\text{th}}$ sample evaluated ($E = 5043$ MPa and $\sigma_y = 153$ MPa) presented a marginal improvement upon both the Young's modulus of sample 49 ($E = 5012$ MPa) and the yield strength of sample 62 ($\sigma_y = 150$ MPa), collapsing the Pareto front to a single Pareto optimal point. Finally, the 68$^{\text{th}}$ sample dominated the 66$^{\text{th}}$ producing the putative globally Pareto optimal solution ($E = 5458$ MPa and $\sigma_y = 165$ MPa), which may be contrasted starkly with the anti-Pareto front comprised of the 14$^{\text{th}}$ ($E = 1114$ MPa and $\sigma_y = 62$ MPa) and 25$^{\text{th}}$ ($E = 813$ MPa and $\sigma_y = 105$ MPa) samples.

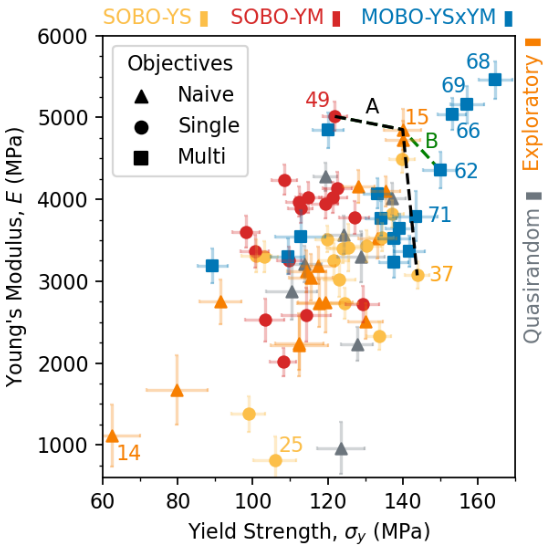

Figure 13: *Results from a multi-Objective Bayesian optimisation routine. Bayesian confidence intervals were a single standard deviation and sampled from the final Gaussian processes. (A) Initial Pareto front as a dashed black line. (B) Pareto front improvement as a dashed green line. YS, Yield strength. YM, Young's modulus. SOBO, Single-objective Bayesian optimisation. MOBO, Multi-objective Bayesian optimisation.*

Figure 14 presents a matrix of slice plots involving the two Gaussian processes as they existed at the culmination of the multi-objective experiment. Every plot holds $^2/_3$ predictors constant whilst the column dictates which of the three predictors is varied. Each row of the matrix conditions the Gaussian process $\mathcal{GP}$ on a different set of predictor constants, these are derived from either the 14$^{\text{th}}$ sample ($\mathbf{x}^{i=14}$), the 68$^{\text{th}}$ sample ($\mathbf{x}^{i=68}$), or the mean of all samples' predictors ($\mathbf{x}^{\mu}$). The graphs are further annotated with vertical lines representing sample predictor values ($\{x_1, x_2, x_3\}$): $\mathbf{x}^{i=14} = \{0.98, 0.5, 0.93\}$ $\mathbf{x}^{i=68} = \{0.72, 0.4, 0.5\}$ $\mathbf{x}^{\mu} = \{0.71, 0.64, 0.64\}$.

Row B of Figure 14 describes the processes as conditioned upon the experimental mean of sample predictors ($\mathcal{GP}|\mathbf{x}^{\mu}$); these graphs provide rough intuition with respect to the objective topography of the parameter space but naturally fail to account for why specific combinations of predictor values (e.g. sample 68 or 14) exhibit higher or lower performance in both the metrics of interest.

In Figure 14A we see the processes conditioned on the 68$^{\text{th}}$ sample's predictors $\mathcal{GP}|\mathbf{x}^{i=68}$: in the case of predictor x_3's yield strength process (Fig.14:A3) we see a topographical deviation from the mean conditioned process (Fig.14:B3). This deviation was a flattening in the region where $x_3 = [0.5, 0.65]$ which shifted the local optimum toward $x_3 = 0.5$: this resulted in an alignment of both local optima of the two processes, which partly explains why marginalisation of UPR content by sample 68 leads to improved performance in both objective metrics.

Although the troughs apparent in the mean processes with respect to x_1 and x_3 (Fig.14:B1&B2) hinted at why the 14$^{\text{th}}$ sample's predictor value combination produced suboptimal performance in both the objective metrics, the x_2 topography failed to do so. On examination of the x_2 topography of the processes conditioned on the 14$^{\text{th}}$ sample's predictors $\mathcal{GP}|\mathbf{x}^{i=14}$ we find large deviations from the mean processes: both the topographies have fallen noticably, the Young's modulus process has flattened, and the yield strength process inverted such that the local optimum shifted from $x_3 = 0.4$ to $x_3 = 1$. Furthemore, this inversion presented a local trough in the vicinity of the 14$^{\text{th}}$ sample explaining further why this sample behaved suboptimally.

Figure 15 presents the strong relation between initiating agent content maximisation and improvements to sample optimality: the average content is 13% for the anti-Pareto front, 44% for the initial Pareto front, and 50% for the putative global optimum. Furthermore, the initiating agent's composition also described a somewhat weaker relation between dibenzoyl peroxide marginalisation and sample optimality: anti-Pareto front 13.3%, initial Pareto front 12.3%, and 8% for the putative global optimum sample. No analagously clear trend existed with respect to the unsaturated polyester resin composition. These three observations are respectively reflected in the Gaussian process topographies presented in Figure 14: when considering predictor x_3 (column 3: controlling balance of unsaturated polyester resin to initiating agent) we see that Gaussian processes clearly describe a negative correlation between x_3 and optimality in all three slice plots (A3,B3,C3); these strong correlations and coinciding local optima between $\mathcal{GP}$'s would explain why there is a clear global link between the balance of unsaturated polyester resin and initiating agent and sample optimality, With respect to x_2 (column 2: controlling initiating agent composition) the $\mathcal{GP}$'s display a negative correlation between x_2 and optimality in two of the slice plots (A2,B2), but this relation disappears and is replaced with a flattened and lowered process in at least one region of the parameter space (C2). In the case of x_1 (column 1: controlling unsaturated polyester resin composition) we can see that the $\mathcal{GP}$'s display a topography with singular peaks at varying positions in all the different slice plots (A1,B1,C1); this added optimisation complexity explains well why there is not obvious global trend relating the unsaturated polyester resin content with sample optimality.

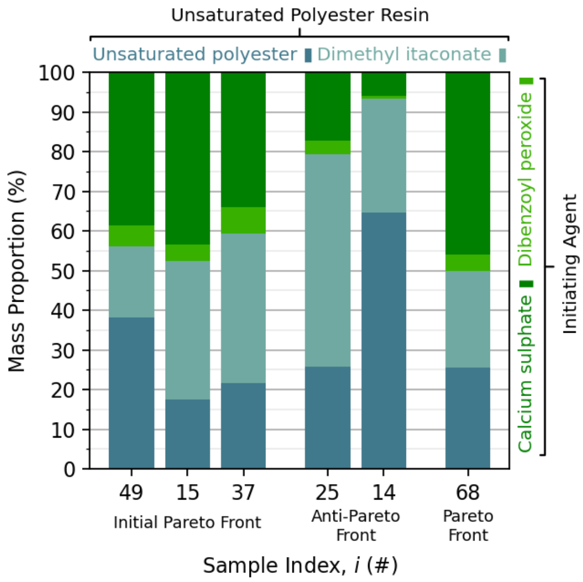

Figure 15: *Compositional breakdown for samples of interest including the initial Pareto front, the final anti-Pareto front, and the final Pareto front.*

4 Conclusion

This work shows that bio-based particulate-reinforced thermosets can be successfully optimised with respect to their compressive yield strength and Young's modulus by adapting their composition during manufacture. Although filler content proved to be the means by which a large degree of strength was obtained, careful trade-off between unsaturated polyester and reactive diluent content was necessary in the high-filler region of the parameter space to obtain the globally optimal results. The detailed experiments show how the multi-objective Bayesian optimisation employing the log noisy expected hypervolume improvement acquisition function was capable of navigating this trade-off, producing a final Pareto front consisting of a single dominating point exhibiting $E = 5,458$ MPa and $\sigma_y = 165$ MPa.

In the domain of a sample-constrained single-objective Bayesian optimisation using the log noisy expected improvement acquisition function, a bio-based composite was obtained which exhibited 69% of the Young's

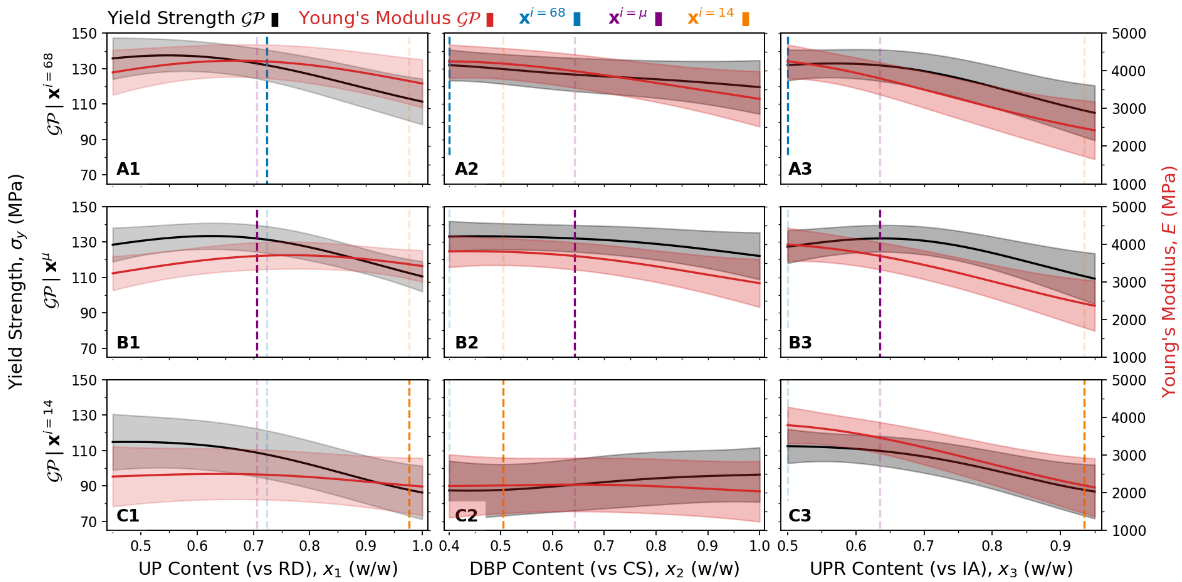

Figure 14: *Slice plot of the Gaussian processes developed by the multi-objective experiment with mean function and a 95% confidence interval* (1.96σ) *presented. UP, Unsaturated polyester. RD, Reactive diluent. DBP, Dibenzoyl peroxide. CS, Calcium sulphate. UPR, Unsaturated polyester resin. IA, Initiating agent.*

modulus of a conventional styrene-based material which had been optimised in the same fashion. Cross-comparison of the aforementioned multi-objective putative global optimum sample with this styrene-based sample developed, showed that it attained 75% the conventional material's strength.

Given the interesting objective topographies witnessed within the parameter space with respect to predictor x_3 which concerns the balance between unsaturated polyester and reactive diluent, future work could examine how reactive diluents impact the mechanical properties of these materials. An optimisation of the styrene-based analogues in the single-objective setting with respect to the objective of yield strength is lacking herein and would be welcomed.

5 Bibliography

[1] Abdeen Ahamad et al. "Synthesis of unsaturated polyester resin- effect of anhydride composition". In: *Designed Polymers and Monomers* 4.3 (2001), pp. 261–268.

[2] Rodrigo Albuquerque et al. "Designing formulations of bio-based, multicomponent epoxy resin systems via machine learning". In: *MRS Bulletin* 49 (2024), pp. 59–70.

[3] Sebastian Ament et al. *Unexpected Improvements to Expected Improvement for Bayesian Optimization.* 2025. arXiv: `2310.20708`. URL: `https://arxiv.org/abs/2310.20708`.

[4] Sharifah Aziz et al. "Modified polyester resins for natural fibre composites". In: *Composites Science and Technology* 65 (2005), pp. 525–535.

[5] Richa Bafana and R Pandey. "New approaches for itaconic acid production-bottlenecks and possible remedies". In: *Critical Reviews in Biotechnology* 38.1 (2017).

[6] P Bamane et al. "Replacement of traditional unsaturated acid by bio-based itaconic acid in the preparation of isophthalic acid-based unsaturated polyester resin". In: *Progress in Organic Coatings* 147.105743 (2020).

[7] J Berlowska et al. “A low-cost method for obtaining high-value bio-based propylene glycol from sugar beet pulp”. In: *RSC Advances* 5 (2015), pp. 2299–2304.

[8] Akos Bodnar et al. “Method of Preparation of Unsaturated Polyester Resins from Maleic Acid or Anhydride”. 4902773. 1990.

[9] Ashish Bohre et al. “Single-step production of bio-based methyl methacrylate from biomass-derived organic acids using solid-catalyst material for cascade decarboxylation-esterification reactions”. In: *ACS Sustainable Chemistry and Engineering* 9.7 (2021).

[10] Chemix. *Chemix - Editor for lab and science diagrams*. Retrieved 4th August 2025. 2025. URL: https://chemix.org.

[11] Huaguo Chen et al. “Recycling thermoset plastic waste for manufacturing green cement mortar”. In: *Cement and Concrete Composites* 137.104922 (2023).

[12] J Chen et al. “Synthesis and application of polyepoxide cardanol glycidyl ether as biobased polyepoxide reactive diluent for epoxy resin”. In: *ACS Sustainable Chemistry and Engineering* 3 (2015), pp. 1164–71.

[13] Benny Cherian and Eby Thachil. “Synthesis of unsaturated polyester resin-effect of choice of reactants and their relative proportions”. In: *International Journal of Polymeric Materials* 53 (2004), pp. 829–845.

[14] C Costa et al. “Going greener- synthesis of fully biobased unsaturated polyesters for styrene crosslinked resins with enhanced thermomechanical properties”. In: *eXPRESS Polymer Letters* 11.11 (2017), pp. 885–898.

[15] John Cowie. “Physical properties of polymers based on itaconic acid”. In: *Pure and Applied Chemistry* 51 (1979), pp. 2331–2343.

[16] John Cowie et al. “Poly(alkyl itaconates)- 4. Glass and sub-glass transitions in the di-alkyl ester series, methyl to hexyl”. In: *Polymer* 18 (1977), pp. 612–616.

[17] K Crombecq et al. “Efficient space-filling and non-collapsing sequential design strategies for simulation-based models”. In: *European Journal of Operational Research* 214 (2011), pp. 683–696.

[18] Jinyue Dai et al. “2,5-Furandicarboxylic acid- and itaconic acid-derived fully biobased unsaturated polyesters and their cross-linked networks”. In: *Industrial and Engineering Chemistry Research* 56 (2017), pp. 2650–2657.

[19] Mohanprasad Dasari et al. “Low-pressure hydrogenolysis of glycerol to propylene glycol”. In: *Applied Catalysis A- General* 281.1-2 (2005), pp. 225–231.

[20] OWID (Our World in Data). *Plastic Pollution*. Ed. by Hannah Ritchie et al. Retrieved July 12th, 2025. 2025. URL: https://ourworldindata.org/plastic-pollution.

[21] Samuel Daulton et al. *Parallel Bayesian Optimization of Multiple Noisy Objectives with Expected Hypervolume Improvement*. Retrieved 1st October 2025. arXiv, 2021. eprint: 2105.08195. URL: https://arxiv.org/abs/2105.08195.

[22] Orla Eriksson et al. “Plastic waste as a fuel - CO2-neutral or not?” In: *Energy and Environmental Science* 2 (2009), pp. 907–914.

[23] L Ettre. “Nomenclature for chromatography”. In: *Pure and Applied Chemistry* 65.4 (1993), pp. 819–872.

[24] Leonhard Euler et al. *Methodus inveniendi lineas curvas maximi minimive proprietate gaudentes, sive, Solutio problematis isoperimetrici latissimo sensu accepti*. Ed. by Leonhard Euler. 1st ed. Apud Marcum-Michaelem Bousquet and Socios, 1744.

[25] Facebook. *Ax*. Version 1.1.2. Sept. 27, 2025. URL: `https://github.com/facebook/Ax`.

[26] Timothy Felthouse et al. "Maleic Anhydride, Maleic Acid, and Fumaric Acid". In: *Kirk-Othmer Encyclopedia of Chemical Technology*. John Wiley and Sons, Ltd, 2001.

[27] Peter Frazier. *A Tutorial on Bayesian Optimization*. Retrieved 28th July 2025. arXiv, 2018. eprint: `1807.02811`. URL: `https://arxiv.org/abs/1807.02811`.

[28] Roman Garnett. *Bayesian Optimization*. Cambridge University Press, 2023.

[29] Roland Geyer et al. "Production, use, and fate of all plastics ever made". In: *Science Advances* 3.e1700782 (2017).

[30] Marianne Gilbert. "Brydson's Plastics Materials". In: ed. by Marianne Gilbert. 8th ed. 2017. Chap. 1.

[31] Sidney Goodman. *Handbook of Thermoset Plastics*. Ed. by Sidney Goodman. 1st ed. Noyes Publications, 1986.

[32] Hiroshan Hettiarachchi et al. "Microplastic Pollution Prevention- The Need for Robust Policy Interventions to Close the Loopholes in Current Waste Management Practices". In: *International Journal of Environmental Research and Public Health* 20.14 (2023).

[33] Thomas Højlund-Dodd. *BO4ACST*. Version 1.0.0. Sept. 27, 2025. URL: `https://github.com/ThomasDodd97/BO4ACST_Ax1`.

[34] Andrew Horwood and Nachiappan Chockalingam. "Clinical Biomechanics in Human Locomotion". In: ed. by Andrew Horwood and Nachiappan Chockalingam. first. 2023. Chap. 2.

[35] Praveenkumara Jagadeesh et al. "Sustainable recycling technologies for thermoplastic polymers and their composites- A review of the state of the art". In: *Polymer Composites* 43.9 (2022), pp. 5831–5862.

[36] Gregory Kharas et al. "Synthesis and characterization of fumarate-based polyesters for use in bioresorbable bone cement composites". In: *Journal of Applied Polymer Science* 66 (1997), pp. 1123–1137.

[37] Y Kissin. "Catagenesis and composition of petroleum- Origin of n-alkanes and isoalkanes in petroleum crudes". In: *Geochimica et Cosmochimica Acta* 51 (1987), pp. 2445–2457.

[38] J Korbar et al. "Process of unsaturated polyester resin synthesis on a laboratory scale". In: *Polymer Engineering and Science* 33.18 (1993), pp. 1212–1216.

[39] John Lawrence. *Polyester Resins*. Ed. by Herbert Simonds. 1st ed. Reinhold Publishing Corporation, 1960.

[40] W Lee et al. "Ethylbenzene dehydrogenation into styrene- kinetic modeling and reactor simulation". In: *Industrial and Engineering Chemistry Research* 47.23 (2008).

[41] Edward Leonard. *Vinyl and Diene Monomers Part 1 (High Polymers XXIV) (Volume 24)*. Wiley, 1970.

[42] Peter Levi and Jonathan Cullen. "Mapping global flows of chemicals- from fossil fuel feedstocks to chemical products". In: *Environmental Science and Technology* 52.4 (2018).

[43] Cheng Li et al. “Rapid Bayesian optimisation for synthesis of short polymer fibre materials”. In: *Scientific Reports* 7.5683 (2017).

[44] Xiukai Li et al. “Selective aerobic oxidation of furfural to maleic anhydride with heterogenous Mo-V-O catalysts”. In: *Green Chemistry* 18 (2016), pp. 2976–2980.

[45] Jingkai Liu et al. “Recent development on bio-based thermosetting resins”. In: *Journal of Polymer Science* 59.14 (2021), pp. 1474–1490.

[46] Yin Lu et al. “Ultrahigh performance benzoxazine/poly(silane arylacetylene) thermosetting composite materials based on smart molecular engineering and machine learning”. In: *Composites Part B- Engineering* 307.112863 (2025).

[47] J Martin. “Materials for Engineering”. In: ed. by J Martin. 3rd ed. 2006. Chap. 2.

[48] Rebekah McKenna and D Nielsen. “Styrene biosynthesis from glucose by engineered E. coli”. In: *Metabolic Engineering* 13.5 (2011), pp. 544–554.

[49] Takuya Minami et al. “Prediction of repeat unit of optimal polymer by Bayesian optimization”. In: *MRS Advances* 4 (2019), pp. 1125–1130.

[50] Taofeeq Moshood et al. “Sustainability of biodegradable plastics- New problem or solution to solve the global plastic pollution?” In: *Current Research in Green and Sustainable Chemistry* 5.100273 (2022).

[51] Taiseer Nafea et al. “Status of management and mitigation of microplastic pollution”. In: *Critical Reviews in Environmental Science and Technology* 54.24 (2024), pp. 1734–1756.

[52] Koichi Nagai. “New developments in the production of methyl methacrylate”. In: *Applied Catalysis A- General* 221.1-2 (2001), pp. 367–377.

[53] Vesna Panic et al. “Simple one-pot synthesis of fully biobased unsaturated polyester resin based on itaconic acid”. In: *Biomacromolecules* 18 (2017), pp. 3881–3891.

[54] Olga Pantic et al. “The effect of glycol derivatives on the properties of bio-based unsaturated polyesters”. In: *Polymers* 14.2970 (2022).

[55] Brian Parkyn. *Polyesters Volume 2 Unsaturated Polyesters and Polyester Plasticisers*. London Iliffe Book Ltd, 1967.

[56] Giulia Pavarelli et al. “A new process for maleic anhydride synthesis from a renewable building block-the gas-phase oxidehydration of bio-1-butanol”. In: *ChemSusChem* 8.13 (2015), pp. 2250–2259.

[57] S Pickering. “Recycling technologies for thermoset composite materials—current status”. In: *Composites Part A- Applied Science and Manufacturing* 37.8 (2006), pp. 1206–1215.

[58] Meta PyTorch. *botorch*. Version 0.15.1. Oct. 5, 2025. URL: `https://github.com/meta-pytorch/botorch`.

[59] Maria Ricciardi et al. “Bio-propylene glycol as value-added product from Epicerol process”. In: *Sustainable Chemistry and Pharmacy* 6 (2017), pp. 10–13.

[60] Florian Rothenhäusler and Holger Ruckdaeschel. “Strategies for the fast optimization of the glass transition temperature of sustainable epoxy resin systems via machine learning”. In: *Journal of Applied Polymer Science* 141.21 (2024).

[61] Caleb Schoenholz and Navid Zobeiry. “An accelerated process optimization method to minimize deformations in composites using theory-guided probabilistic machine learning”. In: *Composites Part A- Applied Science and Manufacturing* 176.107842 (2024).

[62] Ina Schoon et al. "Catalyst influence on undesired side reactions in the polycondensation of fully bio-based polyester itaconates". In: *Polymers* 9.693 (2017).

[63] S Seraji et al. "Fire-retardant unsaturated polyester thermosets- The state-of-the-art, challenges and opportunities". In: *Chemical Engineering Journal* 430.5443 (2022).

[64] Benjamin Shields et al. "Bayesian reaction optimization as a tool for chemical synthesis". In: *Nature* 590 (2021), pp. 89–96.

[65] Jasper Snoek et al. 2012. arXiv: `1206.2944 [stat.ML]`. URL: `https://arxiv.org/abs/1206.2944`.

[66] Lea Sollka and Karen Lienkamp. "Progress in the free and controlled radical homo- and co-polymerisation of itaconic acid derivatives- toward functional polymers with controlled molar mass distribution and architecture". In: *Macromolecular Rapid Communications* 42.4 (2020).

[67] Pavle Spasojevic et al. "Optimization of reactive diluent for bio-based unsaturated polyester resin- a rheological and thermomechanical study". In: *Polymers* 13.2667 (2021).

[68] American Institute of Steel Construction. *Manual of Steel Construction-Allowable Stress Design*. American Institute of Steel Construction, 1989.

[69] Michael Stein. *Interpolation of Spatial Data*. Ed. by Michael Stein. 1st ed. Springer New York, 1999.

[70] Shogo Takasuka et al. "Bayesian optimization of radical polymerization reactions in a flow synthesis system". In: *Science and Technology of Advanced Materials- Methods* 4.1 (2024).

[71] Ryokei Tanaka and Hiroyoshi Iwata. "Bayesian optimization for genomic selection- a method for discovering the best genotype among a large number of candidates". In: *Theoretical and Applied Genetics* 131 (2017), pp. 95–105.

[72] Siew Teong et al. "Hydroxymethylfurfural production from bioresources-past, present and future". In: *Green Chemistry* 16 (2014), pp. 2015–2026.

[73] Eva Topham. "Recycling offshore wind farms at decommissioning stage". In: *Energy Policy* 129 (2019), pp. 698–709.

[74] J Trejbal et al. "Determination of density, viscosity, and saturated vapour pressure of various itaconic acid esters". In: *International Journal of Thermophysics* 43.51 (2022).

[75] Shubham Utekar et al. "Comprehensive study of recycling of thermosetting polymer composites– Driving force, challenges and methods". In: *Composites Part B - Engineering* 207.108596 (2021).

[76] Ankur Vaidya and Kamla Pathak. "Applications of Nanocomposite Materials in Dentistry". In: ed. by Abdullah Asiri et al. first. 2018. Chap. 17.

[77] Jiefeng Xiao et al. "A novel approach for 1,2-propylene glycol production from biomass-derived lactic acid". In: *Process Biochemistry* 50.5 (2015), pp. 793–798.

[78] Xinyao Xu et al. "Efficient exploration of compositional space for high-performance copolymers via Bayesian optimization". In: *Chemical Science* 14 (2015), pp. 10203–10211.

[79] Wenlin Zhao et al. "Design of multicomponent thermosetting polymers with enhanced tensile properties through active learning". In: *Composite Science and Technology* 256.110779 (2024).

A Styrene-Based Experiment

An experiment identical to that of the single-objective Bayesian optimisation for Young's modulus was carried out with the conventional reactive diluent styrene in place of the bio-based dimethyl itaconate. Figure 16 shows an optimisation progression plot for the styrene-based system. The single-objective Bayesian optimisation technique shows strong performance in terms of both sample efficiency and scale of improvement above the baseline. On conclusion of this sequential optimisation, the putative global optimum (sample 18) exhibited a compressive Young's modulus of 7,271 MPa.

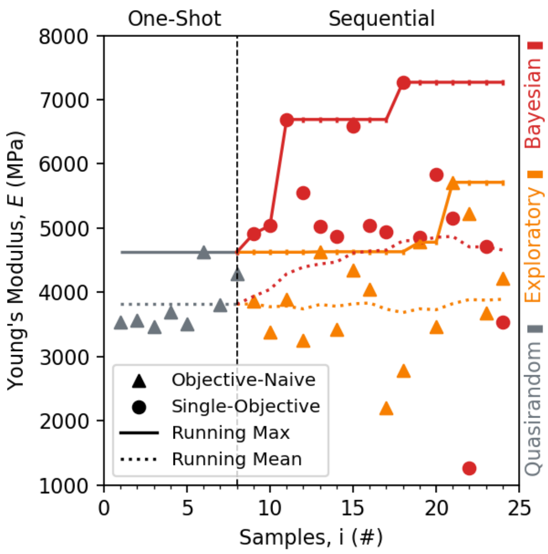

Figure 16: *Progression of a sequential optimisation experiment concerning the maximisation of yield strength with a styrene-based thermoset.*